03 — Volatility across regimes

Scope. All findings describe the visible subset of pharmacy revenue (counter cash receipts, about 65%); see NB01 for the recording basis. 25 December 2019 (an institutional credit-settlement day, per NB01) is excluded throughout.

This notebook asks whether day-to-day revenue variability changed across the four regimes identified in NB02 (pre-COVID, acute crash, partial rebound, lower plateau).

The test compares the variance of log-revenue residuals from a piecewise-mean model against a pooled-null circular block-bootstrap distribution. It reports the statistic |log(σ²_A / σ²_B)|, with signed log ratios (translated to standard-deviation ratios via σ_A/σ_B = exp(½ × log ratio)) as effect sizes and within-regime bootstrap 95% intervals.

These settings were fixed before running the tests below, but the general question ("did volatility change?") had already been explored on this dataset, so the analysis is exploratory.

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from pathlib import Path

from pharmacy_covid_analysis.bootstrap import (
    bootstrap_variance_test,
    bootstrap_log_ratio_ci,
    holm_adjust,
)

sns.set_theme(style="whitegrid", context="notebook")
plt.rcParams["figure.figsize"] = (11, 4)
plt.rcParams["axes.titleweight"] = "bold"
```

In [2]:
```
DATA = Path("../data/daily_sales_final.csv")
df = pd.read_csv(DATA, parse_dates=["date"])
df = df.sort_values("date").set_index("date").asfreq("D")

# Per NB01, 25 December 2019 is an institutional credit-settlement day, not a
# regular trading day. Exclude it here so it does not distort regime means or
# residual variances.
y_raw = df["revenue_scaled"].copy()
y_raw.loc[pd.Timestamp("2019-12-25")] = np.nan
y = y_raw.dropna()

log_y = np.log(y)
dow_means = log_y.groupby(log_y.index.dayofweek).transform("mean")
log_y_dow = log_y - dow_means
```

```
print(
    f"Prepared signal: {len(log_y_dow)} observed days "
    f"from {log_y_dow.index.min().date()} to {log_y_dow.index.max().date
()}"
)
```

```
Prepared signal: 469 observed days from 2019-12-15 to 2021-05-10
```

Regimes and residuals

The four regimes are those identified in NB02 by exact search (Dynp K=3, matched by PELT at pen=10) on the same prepared signal: log revenue with day-of-week means removed, 25 December 2019 excluded. Break dates are 24 March, 28 May and 4 December 2020, with 95% bootstrap CIs of 23–25 March, 26–30 May, and October 2020 – January 2021 respectively. The lower-plateau start is uncertain by a few months; the sensitivity section below revisits this.

Residuals are computed as (log-revenue day-of-week deseasonalised) minus its regime mean. Testing on these residuals rather than raw revenue removes the level shifts and the weekday pattern, so what is compared across regimes is day-to-day variability around each regime's mean and net of the average weekday pattern (a regime-aware version is in the robustness section below).

In [3]:
```
REGIMES = {
    "pre_covid": ("2019-12-15", "2020-03-23"),
    "acute_crash": ("2020-03-24", "2020-05-27"),
    "partial_rebound": ("2020-05-28", "2020-12-03"),
    "lower_plateau": ("2020-12-04", "2021-05-10"),
}
REGIME_LABEL = {
    "pre_covid": "Pre-COVID",
    "acute_crash": "Acute crash",
    "partial_rebound": "Partial rebound",
    "lower_plateau": "Lower plateau",
}


def build_residuals(signal, regimes):
    """Piecewise-mean residuals for a given regime dictionary."""
    label = pd.Series(index=signal.index, dtype="object")
    for name, (start, end) in regimes.items():
        mask = (signal.index >= start) & (signal.index <= end)
        label.loc[mask] = name
    means = signal.groupby(label).mean()
    return signal - label.map(means), label


residuals, regime_label = build_residuals(log_y_dow, REGIMES)

counts = regime_label.value_counts().reindex(REGIMES.keys())
counts.index = [REGIME_LABEL[k] for k in counts.index]
counts.name = "observed_days"
print("Observed days per regime:")
print(counts.to_string())
```

```
Observed days per regime:
Pre-COVID           92
Acute crash         58
Partial rebound    176
Lower plateau      143
```

Two scales for measuring variability

The following table shows day-to-day variability on two scales for each regime: (1) raw revenue standard deviation and coefficient of variation, and (2) log-revenue standard deviation with day-of-week and regime means removed. Revenue fluctuations here are roughly proportional to the level, so raw standard deviations scale with the mean and they are not directly comparable across regimes with different means. Log-scale standard deviations do not: they measure relative day-to-day variability around the regime mean, which is what a "did volatility change" question is usually asking about.

In [4]:
```
# Raw revenue by regime
scale_rows = []
for name, (start, end) in REGIMES.items():
    mask = (y.index >= start) & (y.index <= end)
    raw = y[mask]
    log_res = residuals[regime_label == name].dropna()
    scale_rows.append(
        {
            "regime": REGIME_LABEL[name],
            "n": int(mask.sum()),
            "raw mean (LC/day)": raw.mean(),
            "raw SD (LC/day)": raw.std(ddof=1),
            "raw CV": raw.std(ddof=1) / raw.mean(),
            "log-residual SD": log_res.std(ddof=1),
        }
    )
scale_df = pd.DataFrame(scale_rows)
with pd.option_context("display.float_format", "{:,.4f}".format):
    print(scale_df.to_string(index=False))
```

```
         regime   n  raw mean (LC/day)  raw SD (LC/day)  raw CV  log-resid
ual SD
      Pre-COVID  92       143,105.6427      17,701.9644  0.1237
0.1244
    Acute crash  58        87,209.4145      15,056.4625  0.1726
0.1438
Partial rebound 176       125,588.8737      20,791.9762  0.1656
0.1363
  Lower plateau 143       110,405.6222      17,703.3364  0.1603
0.1336
```

Two observations from this table. First, raw SDs do not fall uniformly after COVID: only the acute-crash regime has a lower raw SD than pre-COVID; partial rebound has a higher raw SD, and the lower plateau is close to pre-COVID. So a simple "raw SD fell" story does not hold across regimes. Second, the raw coefficient of variation (raw SD / mean) is *higher* in every post-COVID regime than pre-COVID, and the log-residual SD is also higher in every post-COVID regime. The formal tests below run on

the log-residual scale, which is the scale most naturally interpreted as "day-to-day variability net of level".

Rolling standard deviation across regimes

In [5]:
```
# 21-day rolling standard deviation of residuals (three weekly cycles; a
# reasonable constant, not tuned to the results). The window counts obse
rved
# days, so a window spanning a missing week covers more than 21 calendar
days.
# min_periods=15 (about two weeks) lets the window start early enough to
see
# the pre-COVID regime.
rolling_std = residuals.rolling(21, min_periods=15).std()

# Reindex to every calendar day so missing weeks appear as gaps rather t
han
# straight-line bridges in the plot.
full_range = pd.date_range(rolling_std.index.min(), rolling_std.index.ma
x(), freq="D")
rolling_std_plot = rolling_std.reindex(full_range)

fig, ax = plt.subplots(figsize=(12, 4.5))

regime_colors = {
    "pre_covid": "#f0f0f0",
    "acute_crash": "#ffe8e8",
    "partial_rebound": "#e8f4ff",
    "lower_plateau": "#f0ede8",
}
for name, (start, end) in REGIMES.items():
    ax.axvspan(
        pd.Timestamp(start),
        pd.Timestamp(end),
        color=regime_colors[name],
        alpha=0.6,
        zorder=0,
    )

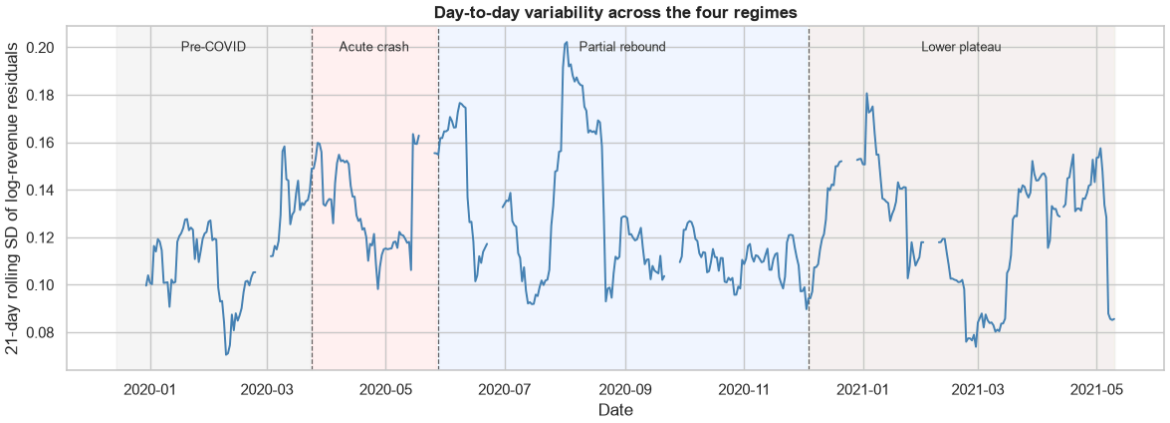
ax.plot(
    rolling_std_plot.index,
    rolling_std_plot.values,
    color="steelblue",
    linewidth=1.4,
    zorder=2,
)

for name, (start, end) in REGIMES.items():
    mid = pd.Timestamp(start) + (pd.Timestamp(end) - pd.Timestamp(star
t)) / 2
    ax.text(
        mid,
        ax.get_ylim()[1] * 0.97,
        REGIME_LABEL[name],
        ha="center",
        va="top",
        fontsize=9,
        color="#333333",
```

```
        zorder=3,
    )

for date in ["2020-03-24", "2020-05-28", "2020-12-04"]:
    ax.axvline(
        pd.Timestamp(date),
        color="black",
        linestyle="--",
        linewidth=0.9,
        alpha=0.6,
        zorder=1,
    )

ax.set_xlabel("Date")
ax.set_ylabel("21-day rolling SD of log-revenue residuals")
ax.set_title("Day-to-day variability across the four regimes")
plt.tight_layout()
plt.savefig("../charts/nb03_rolling_std_by_regime.png", dpi=150, bbox_in
ches="tight")
plt.show()
```



Variability after March 2020 stays close to the pre-COVID band, with several visible peaks (June–July 2020 and January 2021) rather than a sustained fall. The June–July spike coincides with revenue climbing inside the partial-rebound regime, so part of the "variability" it captures is within-regime drift; the robustness check further down uses a within-regime rolling mean to separate short-term noise from drift. The formal tests below quantify whether these curves look different from a single common-variance null.

Pre-specified test settings

Statistic (test): |log(σ²_A / σ²_B)|, two-sided. Ordering the ratio after seeing which regime is more volatile would give an implicitly one-sided test; using the absolute value fixes the direction problem.

Effect size: signed log(σ²_post / σ²_pre); negative values mean lower variability in the post-COVID regime. The 95% percentile interval comes from a *within-regime* block bootstrap (each regime resampled separately), so it quantifies sampling variability of the effect, not evidence against the null. Translation: σ_post / σ_pre = exp(½ × log ratio).

Null and bootstrap (test): pool both regimes' residuals; circular block bootstrap of the pooled residuals; split into original regime sizes; recompute |log(σ²_A / σ²_B)|; p-value = (1 + #{boot ≥ obs}) / (B + 1).

B = 999. Primary block length 14 days (two weekly cycles, matching NB02; a reasonable constant, not tuned to the results); 7 and 28 as sensitivity checks. With 58 observed days in the acute-crash regime, BL=28 gives only about two blocks in that regime and is not informative for crash comparisons: it is reported for the other regimes only.

Primary family: pre-COVID against each post-COVID regime (three tests), with Holm step-down adjustment for family-wise error. The remaining three pairwise comparisons (acute vs rebound, acute vs plateau, rebound vs plateau) are descriptive only.

In [6]:
```
B = 999
BLOCK_LENGTHS = [7, 14, 28]
PRIMARY_BL = 14
SEED = 42
ALPHA = 0.05
```

Primary family: pre-COVID vs each post-COVID regime

In [7]:
```
primary_pairs = [
    ("pre_covid", "acute_crash"),
    ("pre_covid", "partial_rebound"),
    ("pre_covid", "lower_plateau"),
]

primary_rows = []
for i, (name_A, name_B) in enumerate(primary_pairs):
    res_A = residuals[regime_label == name_A].values
    res_B = residuals[regime_label == name_B].values

    obs, _, p_primary = bootstrap_variance_test(
        res_A, res_B, B=B, block_length=PRIMARY_BL, seed=SEED + i
    )
    _, _, p_bl7 = bootstrap_variance_test(
        res_A, res_B, B=B, block_length=7, seed=SEED + 100 + i
    )
    _, _, p_bl28 = bootstrap_variance_test(
        res_A, res_B, B=B, block_length=28, seed=SEED + 200 + i
    )
    # BL=28 not informative for crash comparisons (n_B = 58 days, about
2 blocks at BL=28)
    if name_B == "acute_crash":
        p_bl28_display = "n/a (crash: about 2 blocks at BL=28)"
    else:
        p_bl28_display = f"{p_bl28:.4f}"

    # Effect size: signed log ratio B minus A = post minus pre
    log_ratio_signed, ci_lo, ci_hi, _ = bootstrap_log_ratio_ci(
        res_B, res_A, B=B, block_length=PRIMARY_BL, seed=SEED + 300 + i,
```

```
alpha=ALPHA
    )
    sd_ratio = np.exp(0.5 * log_ratio_signed)
    sd_ratio_lo = np.exp(0.5 * ci_lo)
    sd_ratio_hi = np.exp(0.5 * ci_hi)

    primary_rows.append(
        {
            "comparison": f"{REGIME_LABEL[name_B]} vs {REGIME_LABEL[name
_A]}",
            "|log ratio| (obs)": obs,
            "p (BL=14, primary)": p_primary,
            "p (BL=7)": p_bl7,
            "p (BL=28)": p_bl28_display,
            "signed log ratio": log_ratio_signed,
            "signed log ratio 95% CI": f"[{ci_lo:+.2f}, {ci_hi:+.2f}]",
            "SD ratio (post/pre)": sd_ratio,
            "SD ratio 95% CI": f"[{sd_ratio_lo:.2f}, {sd_ratio_hi:.2
f}]",
        }
    )

# Holm-adjust the three primary p-values
primary_p = [r["p (BL=14, primary)"] for r in primary_rows]
holm_p = holm_adjust(primary_p)
for r, hp in zip(primary_rows, holm_p):
    r["Holm-adjusted p"] = hp

primary_df = pd.DataFrame(primary_rows)
with pd.option_context(
    "display.float_format", "{:.4f}".format, "display.max_colwidth", Non
e
):
    print(primary_df.to_string(index=False))
```

```
                comparison  |log ratio| (obs)  p (BL=14, primary)  p (BL
=7)                          p (BL=28)  signed log ratio signed log rati
o 95% CI  SD ratio (post/pre) SD ratio 95% CI  Holm-adjusted p
    Acute crash vs Pre-COVID             0.2892              0.3270    0.4
080 n/a (crash: about 2 blocks at BL=28)            0.2892          [-0.3
1, +0.85]               1.1556    [0.86, 1.53]           0.9810
Partial rebound vs Pre-COVID             0.1827              0.5360    0.5
140                              0.5330            0.1827          [-0.2
7, +0.70]               1.0957    [0.88, 1.42]           1.0000
  Lower plateau vs Pre-COVID             0.1427              0.5090    0.5
520                              0.5450            0.1427          [-0.2
3, +0.59]               1.0740    [0.89, 1.34]           1.0000
```

SD ratio (post/pre) = exp(½ × signed log ratio): a value of 1.16 reads as post-COVID day-to-day variability about 16% above pre-COVID. The 95% interval on the signed log ratio comes from a within-regime block bootstrap and quantifies sampling variability of the effect size, not evidence against the null.

Reading the results. None of the three primary tests rejects equal variance under Holm adjustment. Point estimates put post-COVID variability slightly above pre-COVID in every comparison, largest in the acute-crash regime (SD ratio 1.16) and smallest in the lower plateau (SD ratio 1.07). The 95% intervals on the effect size

include zero in all three cases, so the data do not distinguish "no change" from a small increase or decrease.

Calibration note. A simulation check (`validation/bootstrap_calibration.py` , run separately: 500 simulated regime pairs per setting at these regime sizes, under an AR(1) null with ϕ = 0 and ϕ = 0.3) gives a false-positive rate of 4.6–6.4% for the test at the 5% level, but only 87–91% coverage for the nominal 95% intervals. The test is well calibrated; the intervals are somewhat too narrow, which changes no conclusion here because they already include zero.

Robustness: separating within-regime drift from short-term noise

Residuals from a flat regime mean count any within-regime drift as variability. The partial-rebound regime contains a visible upward drift in revenue levels (see NB01's daily-revenue chart), and the pre-COVID regime contains an early-March slide, so some of what is counted as "day-to-day variability" is actually within-regime trend. This section reruns the three primary tests on residuals computed differently: instead of subtracting the flat regime mean, the residuals subtract a 15-day centred rolling mean, about two weeks (window truncated at regime boundaries; min_periods = 7 near the ends). The window counts observed days, so near a missing week it spans more calendar days. The resulting residuals measure short-term noise around the local trend within each regime.

In [8]:
```
def detrend_within_regimes(signal, label, window=15):
    """Subtract a centred rolling mean computed separately inside each r
egime,
    so the window does not cross regime boundaries."""
    out = pd.Series(index=signal.index, dtype=float)
    for name in [n for n in label.unique() if not pd.isna(n)]:
        mask = label == name
        sub = signal[mask]
        rm = sub.rolling(window, center=True, min_periods=window // 2).m
ean()
        out.loc[mask] = (sub - rm).values
    return out


detrended = detrend_within_regimes(log_y_dow, regime_label, window=15)

robust_rows = []
for i, (name_A, name_B) in enumerate(primary_pairs):
    rA = detrended[regime_label == name_A].dropna().values
    rB = detrended[regime_label == name_B].dropna().values
    obs, _, p_primary = bootstrap_variance_test(
        rA, rB, B=B, block_length=PRIMARY_BL, seed=SEED + 700 + i
    )
    log_ratio_signed, ci_lo, ci_hi, _ = bootstrap_log_ratio_ci(
        rB, rA, B=B, block_length=PRIMARY_BL, seed=SEED + 800 + i, alpha
=ALPHA
    )
    sd_ratio = np.exp(0.5 * log_ratio_signed)
```

```
    robust_rows.append(
        {
            "comparison": f"{REGIME_LABEL[name_B]} vs {REGIME_LABEL[name
_A]}",
            "|log ratio| (obs)": obs,
            "p (BL=14, primary)": p_primary,
            "signed log ratio": log_ratio_signed,
            "signed log ratio 95% CI": f"[{ci_lo:+.2f}, {ci_hi:+.2f}]",
            "SD ratio (post/pre)": sd_ratio,
        }
    )

robust_p = [r["p (BL=14, primary)"] for r in robust_rows]
robust_holm = holm_adjust(robust_p)
for r, hp in zip(robust_rows, robust_holm):
    r["Holm-adjusted p"] = hp

robust_df = pd.DataFrame(robust_rows)
with pd.option_context(
    "display.float_format", "{:.4f}".format, "display.max_colwidth", Non
e
):
    print(robust_df.to_string(index=False))


# Edge-effect check: drop 7 days at each end of every regime from the de
trended
# residuals and rerun the three primary tests. The centred rolling mean
is
# one-sided within about 7 days of a regime boundary, so this shows how
much
# the detrended-residuals result depends on those edge windows.
def trim_edges_per_regime(series, label, n_trim):
    keep = pd.Series(True, index=series.index)
    for name in [n for n in label.unique() if not pd.isna(n)]:
        mask = label == name
        idx = series.index[mask]
        if len(idx) > 2 * n_trim:
            keep.loc[idx[:n_trim]] = False
            keep.loc[idx[-n_trim:]] = False
    return series[keep], label[keep]


det_trim, lab_trim = trim_edges_per_regime(detrended, regime_label, n_tr
im=7)

edge_rows = []
for i, (name_A, name_B) in enumerate(primary_pairs):
    rA = det_trim[lab_trim == name_A].dropna().values
    rB = det_trim[lab_trim == name_B].dropna().values
    obs, _, p_edge = bootstrap_variance_test(
        rA, rB, B=B, block_length=PRIMARY_BL, seed=SEED + 900 + i
    )
    log_ratio_signed, ci_lo, ci_hi, _ = bootstrap_log_ratio_ci(
        rB, rA, B=B, block_length=PRIMARY_BL, seed=SEED + 1000 + i, alph
a=ALPHA
    )
    sd_ratio = np.exp(0.5 * log_ratio_signed)
    edge_rows.append(
        {
```

```
            "comparison": f"{REGIME_LABEL[name_B]} vs {REGIME_LABEL[name
_A]}",
            "n_A (trimmed)": int((lab_trim == name_A).sum()),
            "n_B (trimmed)": int((lab_trim == name_B).sum()),
            "|log ratio| (obs)": obs,
            "p (BL=14)": p_edge,
            "signed log ratio": log_ratio_signed,
            "signed log ratio 95% CI": f"[{ci_lo:+.2f}, {ci_hi:+.2f}]",
            "SD ratio (post/pre)": sd_ratio,
        }
    )

edge_p = [r["p (BL=14)"] for r in edge_rows]
edge_holm = holm_adjust(edge_p)
for r, hp in zip(edge_rows, edge_holm):
    r["Holm-adjusted p"] = hp

edge_df = pd.DataFrame(edge_rows)
print()
print("Edge-trimmed detrended residuals (drop 7 days each end per regim
e):")
with pd.option_context(
    "display.float_format", "{:.4f}".format, "display.max_colwidth", Non
e
):
    print(edge_df.to_string(index=False))
```

```
                comparison  |log ratio| (obs)  p (BL=14, primary)  signe
d log ratio signed log ratio 95% CI  SD ratio (post/pre)  Holm-adjusted p
   Acute crash vs Pre-COVID             0.5309              0.1110
0.5309          [-0.12, +1.12]              1.3040           0.3330
Partial rebound vs Pre-COVID             0.0095              0.9600
0.0095          [-0.38, +0.43]              1.0047           0.9600
  Lower plateau vs Pre-COVID             0.2266              0.3200
0.2266          [-0.23, +0.64]              1.1199           0.6400
Edge-trimmed detrended residuals (drop 7 days each end per regime):
                comparison  n_A (trimmed)  n_B (trimmed)  |log ratio| (o
bs)  p (BL=14)  signed log ratio signed log ratio 95% CI  SD ratio (post/p
re)  Holm-adjusted p
   Acute crash vs Pre-COVID             78             44             0.1
923     0.5600           0.1923          [-0.31, +0.64]              1.1
009          1.0000
Partial rebound vs Pre-COVID             78            162             0.1
423     0.5820           0.1423          [-0.27, +0.60]              1.0
737          1.0000
  Lower plateau vs Pre-COVID             78            129             0.3
559     0.1820           0.3559          [-0.02, +0.77]              1.1
948          0.5460
```

Same qualitative conclusion: no Holm-adjusted p reaches 0.05. Two observations from the flat-mean-detrended table. First, the partial-rebound comparison now has a SD ratio of 1.00: with the within-regime trend removed, the rebound looks indistinguishable from pre-COVID in day-to-day noise, so the earlier point-estimate excess in the primary table (SD ratio 1.10) was reflecting the rebound's upward drift rather than a change in noise. Second, the acute-crash SD ratio grows from 1.16 to 1.30, because pre-COVID also had a mild downward drift in March that was contributing to its flat-mean residual SD; when that drift is removed, pre-COVID looks quieter and the crash regime looks relatively noisier.

Edge-effect check. The centred 15-day rolling mean is one-sided within about 7 days of each regime boundary, and the acute crash (58 days) has proportionally more edge observations than the longer regimes. Dropping 7 days at each end of every regime before recomputing the tests gives SD ratios of 1.10 (crash), 1.07 (rebound), 1.20 (plateau): the crash figure moves back towards 1 while the plateau moves up. The qualitative conclusion is unchanged: no Holm-adjusted p reaches 0.05, and every effect-size interval includes zero, the plateau's only just (−0.02 to +0.77).

Sensitivity: plateau start moved to CI endpoints

NB02's 95% bootstrap CI for the third break spans October 2020 to January 2021. The primary pre-COVID-vs-lower-plateau test above uses 4 December 2020, the observed break date. This section repeats the test with the plateau start moved to each CI endpoint, so the partial-rebound end date shifts with it. Pre-COVID and acute-crash regimes are unchanged.

In [9]:
```
plateau_starts = [
    ("2020-10-08", "CI lower endpoint", SEED + 400),
    (
        "2020-12-04",
        "Primary (observed)",
        SEED + 2,
    ),  # reuse primary seed for pre_covid vs lower_plateau
    ("2021-01-18", "CI upper endpoint", SEED + 402),
]
plateau_ci_starts = [SEED + 500, SEED + 300 + 2, SEED + 502]

sens_rows = []
for i, ((start_date, label, test_seed), ci_seed) in enumerate(
    zip(plateau_starts, plateau_ci_starts)
):
    # Shifted regime dictionary: rebound ends the day before the new pla
teau start
    shifted = dict(REGIMES)
    shifted["partial_rebound"] = (
        shifted["partial_rebound"][0],
        (pd.Timestamp(start_date) - pd.Timedelta(1, "D")).strftime("%Y-%
m-%d"),
    )
    shifted["lower_plateau"] = (start_date, shifted["lower_plateau"][1])

    residuals_s, label_s = build_residuals(log_y_dow, shifted)
    res_A = residuals_s[label_s == "pre_covid"].values
    res_B = residuals_s[label_s == "lower_plateau"].values

    obs, _, p_primary = bootstrap_variance_test(
        res_A, res_B, B=B, block_length=PRIMARY_BL, seed=test_seed
    )
    log_ratio_signed, ci_lo, ci_hi, _ = bootstrap_log_ratio_ci(
        res_B, res_A, B=B, block_length=PRIMARY_BL, seed=ci_seed, alpha=
ALPHA
    )
    sd_ratio = np.exp(0.5 * log_ratio_signed)
```

```
    sens_rows.append(
        {
            "plateau start": start_date,
            "label": label,
            "|log ratio| (obs)": obs,
            "p (BL=14)": p_primary,
            "signed log ratio": log_ratio_signed,
            "signed log ratio 95% CI": f"[{ci_lo:+.2f}, {ci_hi:+.2f}]",
            "SD ratio (plateau/pre)": sd_ratio,
        }
    )

sens_df = pd.DataFrame(sens_rows)
with pd.option_context(
    "display.float_format", "{:.4f}".format, "display.max_colwidth", Non
e
):
    print(sens_df.to_string(index=False))
```

```
plateau start              label  |log ratio| (obs)  p (BL=14)  signed log
ratio signed log ratio 95% CI  SD ratio (plateau/pre)
   2020-10-08  CI lower endpoint             0.3009     0.1290
0.3009          [-0.05, +0.76]                1.1624
   2020-12-04 Primary (observed)             0.1427     0.5090
0.1427          [-0.23, +0.59]                1.0740
   2021-01-18  CI upper endpoint             0.0559     0.8260
0.0559          [-0.48, +0.54]                1.0284
```

The primary row (4 December 2020) reproduces the pre-COVID-vs-lower-plateau line from the primary table exactly (same seeds). No plateau-start choice gives a raw p-value below 0.05 (values are 0.13, 0.51 and 0.83 for the early, primary and late endpoints respectively), and the SD ratio stays above 1 throughout. The early endpoint (8 October 2020) comes closest to producing a rejection: its p-value of 0.13 is the smallest of the three and its point estimate the largest. This is consistent with the plateau's start being less clearly identified than the first two breaks; it is not a rejection.

Descriptive-only comparisons (not part of the primary family)

The three within-post-COVID pairwise comparisons are reported without formal testing. They are outside the primary family and are reported descriptively.

In [10]:
```
descriptive_pairs = [
    ("acute_crash", "partial_rebound"),
    ("acute_crash", "lower_plateau"),
    ("partial_rebound", "lower_plateau"),
]

desc_rows = []
for i, (name_A, name_B) in enumerate(descriptive_pairs):
    res_A = residuals[regime_label == name_A].values
    res_B = residuals[regime_label == name_B].values
    log_ratio_signed, ci_lo, ci_hi, _ = bootstrap_log_ratio_ci(
        res_A, res_B, B=B, block_length=PRIMARY_BL, seed=SEED + 600 + i,
```

```
alpha=ALPHA
    )
    sd_ratio = np.exp(0.5 * log_ratio_signed)
    desc_rows.append(
        {
            "comparison": f"{REGIME_LABEL[name_A]} vs {REGIME_LABEL[name
_B]}",
            "signed log ratio": log_ratio_signed,
            "signed log ratio 95% CI": f"[{ci_lo:+.2f}, {ci_hi:+.2f}]",
            "SD ratio (A/B)": sd_ratio,
        }
    )

desc_df = pd.DataFrame(desc_rows)
with pd.option_context(
    "display.float_format", "{:.4f}".format, "display.max_colwidth", Non
e
):
    print(desc_df.to_string(index=False))
```

```
                    comparison  signed log ratio signed log ratio 95% CI
SD ratio (A/B)
  Acute crash vs Partial rebound            0.1065          [-0.55, +0.67]
1.0547
    Acute crash vs Lower plateau            0.1465          [-0.47, +0.69]
1.0760
Partial rebound vs Lower plateau            0.0400          [-0.39, +0.51]
1.0202
```

Comparisons involving the acute-crash regime carry the same short-sample caveat as in the primary table: with about four blocks at BL=14, the CIs on those effect sizes are rough.

Robustness: regime-aware day-of-week correction

NB02's day-of-week deseasonalisation used a single pooled day-of-week mean, subtracted uniformly across the sample. NB04 finds the weekly pattern was not stable (details in NB04). This section reruns the three primary comparisons on residuals from a *regime-aware* correction: subtract each observation's regime × day-of-week cell mean rather than a pooled day-of-week mean.

In [11]:
```
# Regime-aware residuals: subtract the regime × day-of-week cell mean di
rectly.
regime_idx = pd.Series(
    sum(
        (log_y_dow.index >= pd.Timestamp(b)).astype(int)
        for b in ["2020-03-24", "2020-05-28", "2020-12-04"]
    ),
    index=log_y_dow.index,
)
# log_y already excludes 25 Dec 2019.
cell_mean = log_y.groupby([regime_idx, log_y.index.dayofweek]).transform
("mean")
residuals_aware = log_y - cell_mean

aware_rows = []
```

```
for i, (name_A, name_B) in enumerate(primary_pairs):
    rA = residuals_aware[regime_label == name_A].dropna().values
    rB = residuals_aware[regime_label == name_B].dropna().values
    obs, _, p_primary = bootstrap_variance_test(
        rA, rB, B=B, block_length=PRIMARY_BL, seed=SEED + 1100 + i
    )
    log_ratio_signed, ci_lo, ci_hi, _ = bootstrap_log_ratio_ci(
        rB, rA, B=B, block_length=PRIMARY_BL, seed=SEED + 1200 + i, alph
a=ALPHA
    )
    sd_ratio = np.exp(0.5 * log_ratio_signed)
    aware_rows.append(
        {
            "comparison": f"{REGIME_LABEL[name_B]} vs {REGIME_LABEL[name
_A]}",
            "|log ratio| (obs)": obs,
            "p (BL=14, primary)": p_primary,
            "signed log ratio": log_ratio_signed,
            "signed log ratio 95% CI": f"[{ci_lo:+.2f}, {ci_hi:+.2f}]",
            "SD ratio (post/pre)": sd_ratio,
        }
    )
aware_p = [r["p (BL=14, primary)"] for r in aware_rows]
aware_holm = holm_adjust(aware_p)
for r, hp in zip(aware_rows, aware_holm):
    r["Holm-adjusted p"] = hp
aware_df = pd.DataFrame(aware_rows)
print("Regime-aware residuals — primary tests (B=999, BL=14):")
with pd.option_context(
    "display.float_format", "{:.4f}".format, "display.max_colwidth", Non
e
):
    print(aware_df.to_string(index=False))
```

```
Regime-aware residuals — primary tests (B=999, BL=14):
                comparison  |log ratio| (obs)  p (BL=14, primary)  signe
d log ratio signed log ratio 95% CI  SD ratio (post/pre)  Holm-adjusted p
   Acute crash vs Pre-COVID             0.3151              0.3000
0.3151          [-0.23, +0.91]              1.1706              0.7740
Partial rebound vs Pre-COVID             0.3107              0.2650
0.3107          [-0.13, +0.87]              1.1680              0.7740
  Lower plateau vs Pre-COVID             0.2702              0.2580
0.2702          [-0.15, +0.73]              1.1446              0.7740
```

Under the regime-aware correction, SD ratios (post/pre) shift up:

- Acute crash: 1.16 → about 1.17 (small change; the crash is the acute-Friday-deepening regime, so both numerator and denominator shrink)
- Partial rebound: 1.10 → about 1.17
- Lower plateau: 1.07 → about 1.14

The ratios imply that the pre-COVID and acute-crash residual SDs shrink more than the rebound and plateau SDs. Part of the reduction is mechanical: the regime-aware correction fits 28 cell means in-sample rather than 7 + 4, and the crash has about eight observations per cell. One explanation (developed in NB04's Friday × regime finding, next notebook) is that their true Friday effects diverge most from the pooled Friday mean, so replacing the single pooled Friday mean with regime-specific ones

removes more spurious variance in those two regimes. The pre-COVID SD sits in the denominator of each SD ratio, so its disproportionate shrinkage pushes the ratios up.

Bootstrap p-values fall from 0.51–0.54 (rebound and plateau, pooled) to about 0.26 (regime-aware), but no Holm-adjusted p reaches 0.05. Under the regime-varying weekly pattern NB04 identifies, the regime-aware residuals measure variability around each regime's own weekly pattern, which is the more accurate version of the question.

Summary

All findings describe counter cash receipts (about 65% of revenue; see NB01), analysed as log revenue with day-of-week means removed and 25 December 2019 excluded. Volatility is measured on residuals from a piecewise-mean model with regime boundaries taken from NB02 (24 March, 28 May and 4 December 2020), so what is compared across regimes is variability around each regime's mean, net of level shifts and the weekday pattern.

Result. No formal evidence of change. A two-sided block-bootstrap test (statistic $|\log(\sigma^2_A / \sigma^2_B)|$; pooled null; circular block bootstrap of the pooled residuals; B = 999; primary block length 14 days) fails to reject equal variance in each of the three primary comparisons (pre-COVID vs each post-COVID regime), with Holm-adjusted p-values ranging from 0.98 to 1.00.

Effect sizes. SD ratios (post/pre) around each regime mean are 1.16 (acute crash), 1.10 (partial rebound) and 1.07 (lower plateau): point estimates slightly above pre-COVID rather than below. The 95% bootstrap intervals on the signed log ratio include zero in every comparison, so the data are consistent with no change, a small increase, or a small decrease.

Why raw SDs can mislead. The scale table shows the acute-crash raw SD is lower than pre-COVID, but mostly because the mean dropped, not because day-to-day scatter shrank. Testing on the log-residual scale strips out this level dependence; that is what "did volatility change" most naturally means, and what the formal test reports.

Robustness to within-regime drift. Removing a 15-day within-regime rolling mean (which strips within-regime trend) gives the same qualitative conclusion. The crash SD ratio depends materially on how its first and last week are handled (1.10 with edge trimming, 1.30 without), and no version rejects at Holm-adjusted $\alpha = 0.05$.

Robustness to the plateau boundary. NB02's 95% CI for the third break spans 8 October 2020 to 18 January 2021. Repeating the pre-COVID-vs-lower-plateau test at both endpoints gives raw p-values of 0.13, 0.51 and 0.83 for the early, primary and late endpoints; the SD ratio stays above 1 throughout. No endpoint produces a rejection at conventional levels.

Descriptive within-post-COVID pairs. The three pairwise comparisons among acute crash, partial rebound and lower plateau show small effect-size point estimates (SD ratios all within 1.02–1.08) with wide intervals that include zero. No formal test is claimed for these; they are reported descriptively.

Calibration. The simulation check in `validation/bootstrap_calibration.py` (see the calibration note above) finds the test well calibrated and the intervals somewhat too narrow, which changes no conclusion here.

Regime-aware weekday correction (NB04 follow-through). NB04 (next notebook) finds the Friday effect changed across regimes, so the pooled weekday correction used in the primary tests leaves regime-specific weekly structure in the residuals. Repeating the primary tests with a regime-aware correction, subtracting the full regime × day-of-week cell mean, pushes the SD ratios to 1.17, 1.17 and 1.14 for acute crash, partial rebound and lower plateau versus pre-COVID. Holm-adjusted p-values are 0.77 across the family, so the conclusion is unchanged.

Overall. The most consistent signal across every version is a slightly higher lower-plateau SD ratio: 1.07 under primary, 1.12 under flat-mean detrending, 1.20 under edge-trimmed detrending, 1.16 at the early plateau-start endpoint, and 1.14 under the regime-aware correction. Every one of these has an interval that includes zero, and none rejects (raw $p \geq 0.13$; Holm-adjusted where a family was defined). The lower-plateau ratio is above 1 in every specification and never significantly so. The pre-COVID vs acute crash and partial rebound comparisons vary substantially with the specification and are best summarised as "no clear signal in either direction".